HealthConnect Final Data Science Outcome

Week 8 - Final Integration, Presentation & Project Showcase

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Final decision

The final documented admission model is reproducible and leakage-safer, but its predictive performance remains close to chance. It is NOT recommended for clinical or operational deployment.

From Week 6 to Week 8

The project evolved from model development to evidence-based final selection.

Week 6

Advanced development

Admission candidate ROC-AUC ≈ 0.497

Week 7

Testing & refinement

Weaknesses, leakage risk and unstable refinements identified

Week 8

Final integration

Final candidate documented; deployment rejected

Week 8 purpose

Select the most defensible final model outcome, integrate validated cross-track findings, document limitations and risks, and communicate the final decision professionally.

Data, target and prediction point

The final modelling question is constrained by what the supplied Rwanda data can actually support.

Integrated records	Admissions	Non-admissions	Attendance target
5,000	586	4,414	Unavailable
appointment-level	11.72% positive	88.28% negative	no no-show field

Actual target

Hospital admission status (`Admitted`) derived from the Admissions sheet and linked by AppointmentID. The target is actual admission, not a proxy.

Final prediction point

Patient arrival / before initial clinical assessment. This prevents the final model from depending on information generated after the decision point.

Key integrity rule: no attendance/no-show target was fabricated.

Cross-track validation changed the model

Data Analytics findings were used as model-design evidence, not just as a discussion record.

Output received from Data Analytics

- Gender, chronic condition and visit type supported
- Age should be calculated at appointment date
- Facility/department IDs require caveats
- Diagnosis/Severity may leak prediction-time information
- Suggested PriorVisits, month and weekday
- Attendance/no-show cannot be validated

Data Science action

- Fixed prediction point at patient arrival
- Recalculated AgeAtAppointment
- Added time-safe PriorVisits
- Added AppointmentMonth and Weekday
- Removed Diagnosis and Severity
- Excluded unvalidated BMI/demographic variables
- Retained FacilityID/DepartmentID with caveats

What changed

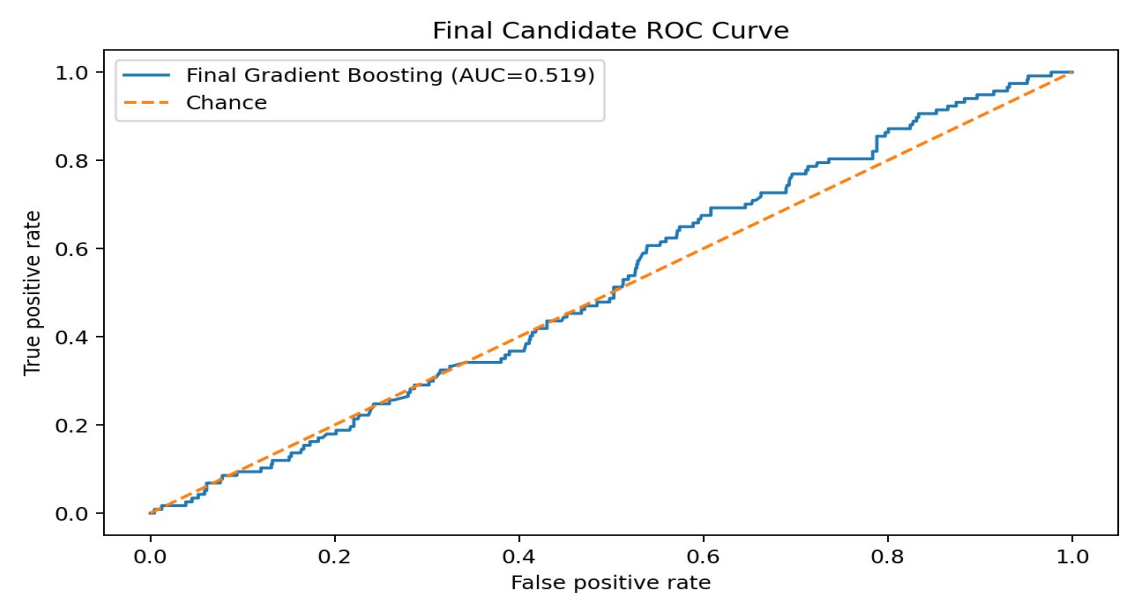
Feature governance improved and leakage risk was reduced. The re-test became more operationally defensible, but performance remained too weak for deployment.

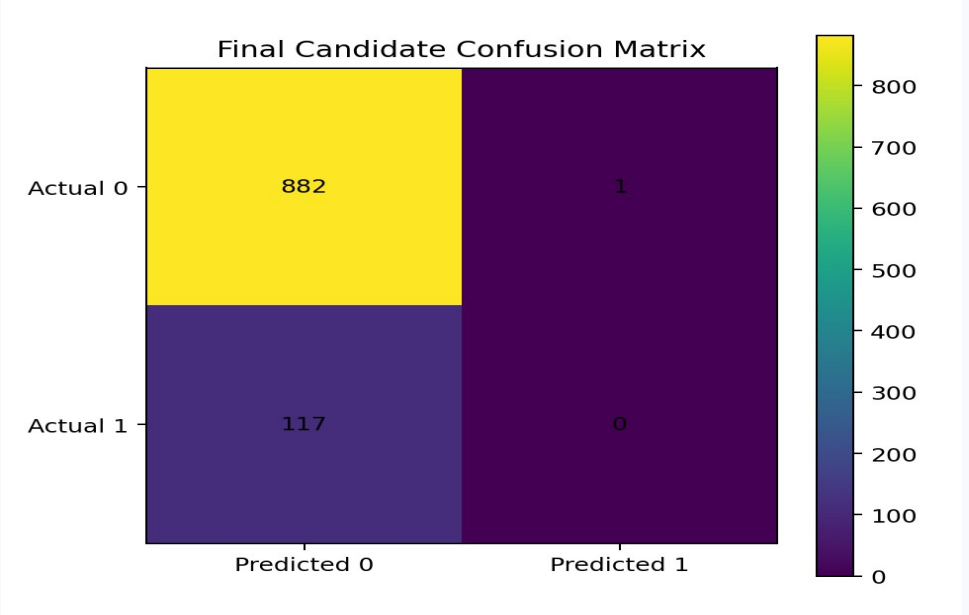
Test → Finding → Action → Re-test → Final decision

Final model evaluation

Gradient Boosting is the final documented candidate - not a deployment-approved model.

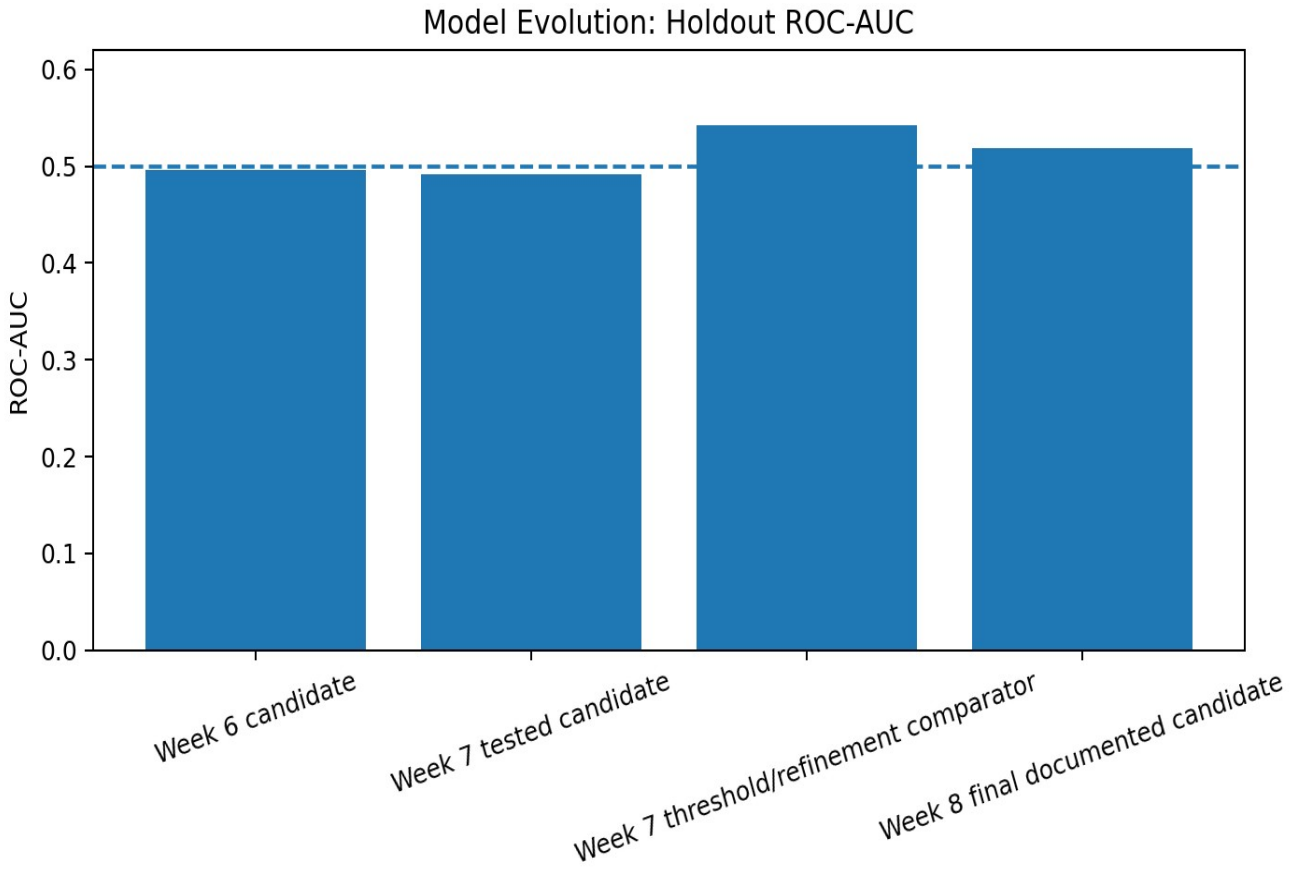
Holdout ROC-AUC	5-fold CV ROC-AUC	Holdout PR-AUC	Recall @ 0.50	Accuracy
0.5189	**0.5237**	**0.1199**	**0.000**	**0.882**
slightly above chance	± 0.0246	positive rate ≈ 0.117	missed all positives	misleading alone





Model evolution and final selection

A better-looking single split was rejected when it did not generalize.



Week 6 candidate

ROC-AUC ≈ 0.497 - near chance.

Week 7 refinement

Holdout ROC-AUC ≈ 0.542 looked better, but 5-fold CV ROC-AUC ≈ 0.487. The apparent gain was rejected as unstable.

Week 8 final documented candidate

Holdout ROC-AUC 0.519; CV ROC-AUC 0.524. Methodologically safer, still not operationally useful.

What the model can - and cannot - do

Final model documentation separates reproducibility value from operational suitability.

Can do

- Reproduce the final feature-engineering and preprocessing workflow
- Demonstrate leakage-aware model validation
- Provide a documented admission-risk experiment
- Support portfolio discussion of responsible model selection
- Provide technical schema requirements for an ML pipeline demonstration

Cannot do

- Reliably predict admission
- Support clinical triage or patient prioritization
- Drive staffing/resource allocation
- Automate patient-level care decisions
- Predict attendance/no-show from the supplied dataset
- Be described as production-ready or clinically validated

Final business decision: DO NOT DEPLOY

How Data Science fits the final HealthConnect solution

The value of the Data Science contribution is the validated decision, not a forced production model.

Data Analytics	Validated findings and feature/timing evidence
Data Science	Tests predictive value, controls leakage, documents model suitability
ML Engineering	Can use schema/preprocessing for technical reproducibility only
Generative AI	Supports administrative interaction within approved knowledge base
Project Management	Coordinates readiness, dependencies and presentation

Integration outcome: the wider solution is protected from relying on an unreliable predictive component.

Recommendations and closing

Responsible Data Science means following the evidence - including negative results.

- Collect an explicit attendance/no-show outcome if missed appointments remain the core use case.
- Obtain stronger predictors that are genuinely available before clinical assessment.
- Freeze a fresh untouched test set before future model selection or threshold tuning.
- Define the operational costs of false negatives and false positives before selecting a threshold.
- Revisit deployment only after materially stronger and stable out-of-sample performance.

Key takeaway

The final model is not a failed project outcome. The validated do-not-deploy decision prevents weak predictions from becoming part of the HealthConnect workflow.

Evidence > appearance of success

Thank you